ARTIFICIAL INTELLIGENCE

Smart Library Book Finder

Intelligent Agent Formulation and PEAS Evaluation

STUDENT	FATHER NAME	SEMESTER	SECTION	ROLL NO.
Ibrahim Khan Jagwal	Khalil ur Rehman Khan	6th	C	23122107

1 Agent Overview

The Smart Library Book Finder is a simple intelligent agent that helps a student find useful books without manually checking the complete library catalog. The user provides a title, author, subject, or topic. The agent checks available records, finds close matches, and returns the most relevant books.

Goal of the agent: Find relevant books quickly and show whether they are available.

2 Graphical Representation

ENVIRONMENT: Library catalog + availability records + user request

Input -> Perception -> Decision / Search -> Output

3 Agent Formulation

- **Environment:** Library database, book records, availability status, categories, and the student using the system.
- **Sensors:** Book title, author name, subject, topic, category, and other search words entered by the user.
- **Internal Processing:** Searches the catalog, compares the query with book records, filters unavailable or weak matches, and ranks useful results.
- **Actuators:** Displays recommended books, availability status, and shelf or location information when available.

PEAS and Environment Analysis

4 PEAS Framework

Element	Evaluation for Smart Library Agent
P - Performance Measure	Relevant recommendations, correct availability information, quick search response, and useful results for the student.
E - Environment	Library users, catalog/database, book records, subject categories, availability status, and search interface.
A - Actuators	Displays ranked books, availability, and location or shelf information.
S - Sensors	User-entered title, author, topic, subject, category, and search keywords.

5 Environment Type

Property	Type	Reason
Observability	Partially Observable	The catalog is visible to the agent, but the user may not state every preference or intention.
Determinism	Deterministic	With the same query and unchanged records, the same search rules produce the same results.
Episodes	Episodic	Each search request can be handled independently from previous searches.
Change	Dynamic	Availability changes when books are borrowed, returned, added, or removed.
Structure	Discrete	Books, categories, and availability states are stored as separate records and values.
Agents	Single Agent	One library agent performs the search and recommendation task for the user.

6 Example Interaction

User query: "Python programming for beginners"

Agent output: Ranks matching beginner Python books and shows which copies are available.